Learning to Contest: Decentralized Robust Fairness in Cooperative MARL via Cross-Attention

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***Abstract*—Fair cooperative multi-agent reinforcement learning (MARL) teams that maximize an egalitarian welfare are exploitable: a single self-interested agent free-rides on the surplus that fair agents forgo to raise the worst-off, and the known remedy is a *centralized* need-based allocator. We show that a decentralized defense becomes possible once contention is *graded*: when a contested resource still delivers a fraction $1-c$, a worst-off cooperator that contests a free-rider strictly improves on yielding (Prop. 1), so leverage exists for every $c < 1$. We introduce CAN, a permutation-equivariant cross-attention policy over agents' observed behaviour that infers how many free-riders are present and responds proportionally—turn-taking when none, contesting just enough when some. Trained against an adversarial league, CAN keeps best-response exploitability near the centralized oracle ($\rho{\approx}1.2$–1.5 vs. $\rho{=}N$ unprotected) at essentially no efficiency cost, where the fair-MARL learners (GGF, FEN, SOTO) each collapse to an exploitable or wasteful extreme. Giving those objectives CAN's identical adversarial training does not rescue them, so the objective—not adversarial training alone—is what makes hardening possible. Against a *committed* (non-adaptive) defector, every learned defense including ours provides deterrence rather than immunity, weakening as the leverage $(1-c)/2$ vanishes. Across further environments and team sizes the same principle sets the scope: robustness holds exactly as far as the game's contest leverage reaches, and we map that boundary rather than claim to remove it.**

***Keywords*—multi-agent reinforcement learning, fairness, exploitability, cross-attention, decentralized execution, league training**

I. Introduction

Cooperative multi-agent reinforcement learning (MARL) increasingly optimizes not only efficiency but *fairness*: methods such as FEN [1] and SOTO [2] maximize a fairness-inducing welfare (typically the Generalized Gini Welfare [3]) over agents' returns so that no agent is starved. These methods assume every agent cooperates in being fair. When one does not—one stakeholder upgrades a controller to minimize its own delay, one client maximizes its own throughput—the fair team is exploitable: because team-oriented agents *sacrifice* their own utility to raise the worst-off, a self-interested agent free-rides on that sacrifice [4]. In a strictly rivalrous (all-or-nothing) resource—where a contested unit is either won by one agent or wasted—this exploit is hard to defend against *at the policy level*: a cooperator that contests a free-rider merely causes a collision, gaining nothing, so a welfare-fair team is indifferent between yielding and contesting; a recent study formalizes this futility and the centralized need-based remedy that sidesteps it [5]. The robust remedy is then to take allocation out of the agents' hands entirely, via a *centralized* need-based mechanism. This paper takes up the harder question that centralization sidesteps: *can decentralized policies—acting only on local, observed information—retain fairness under a self-interested agent?*

We answer affirmatively, with caveats, in three steps.

(1) Leverage exists once contention is graded. The policy-level futility above hinges on *all-or-nothing* contention: a contested resource is either won by one agent or wasted, so a cooperator that contests a free-rider gains nothing. We introduce a *graded-contention* game in which $m \geq 2$ claimers split a fraction $1-c$ of the resource (waste c). We prove (Prop. 1) that for any $c < 1$ a worst-off cooperator that contests a lone free-rider receives $(1-c)/2 > 0$ instead of 0, so contesting *strictly dominates* yielding: decentralized leverage reappears. All-or-nothing is the limiting case $c{=}1$.

(2) The residual difficulty is coordination under uncertainty. Leverage is not a policy. The number of free-riders D is unknown and varies episode to episode (including $D{=}0$). A fixed rule cannot win: always-contest pays the waste c even when nobody defects, and always-yield collapses the moment someone does. A robust cooperator must *infer* D from observed behaviour and respond *proportionally*—do nothing (turn-take) when $D{=}0$, contest just enough when $D \geq 1$.

(3) A cross-attention policy, trained against a league, realizes it. We introduce **CAN** (Cross-Attention Networks), a shared, permutation-equivariant policy in which each agent attends over the *observed behaviour* of all agents (utilities, claim-rates, who is worst-off) to estimate contention and act. What is load-bearing is *behaviour-conditioning with adaptive aggregation*—the response must depend on *how many* others are grabbing, not on fixed identities—and among permutation-respecting aggregators cross-attention is the most stable instance (Sec. V-H). We train CAN against an adversarial league (PSRO [6]) and audit it with both a fresh best-response defector and a committed always-claim script. CAN keeps trained-best-response exploitability low ($\rho{\approx}1.2$–1.5) across all contention levels while wasting essentially nothing when no free-rider is present (efficiency ≈ 1.0), approaching the centralized oracle without any central allocator; against the *committed* defector its robustness is a deterrence effect that decays as leverage vanishes (Sec. V-E).

We are deliberate about limits. We report a negative-leaning result—single adversary co-training is unreliable at low contention—and two genuine fragilities: zero-shot transfer of an $N{=}6$ policy to teams of 12 and 24 degrades at high contention, and robustness to a *committed* (rather than adaptive) defector decays as leverage vanishes, for every method including ours (Sec. V-E). The contribution is a controlled, honest map of how far decentralized robust fairness reaches, not a claim that the decentralization gap is closed.

II. Background and Related Work

Welfare-based fair MARL. Given per-agent returns $\mathbf{u} = (u_1, \ldots, u_N)$, fair-MARL maximizes a Schur-concave welfare $W(\mathbf{u})$ that is increasing (efficiency) and favours equity, e.g. the Generalized Gini Welfare $W(\mathbf{u}) = \sum_k w_k u_k^{\uparrow}$ with decreasing weights w_k so the worst-off agent dominates [3]. FEN [1] and SOTO [2] are representative learners. These objectives create the appropriable surplus we must defend.

Exploitability and the decentralization gap. The vulnerability is a fairness-specific instance of free-riding in sequential social dilemmas [7]; that welfare-maximizing cooperation is gameable in general was noted by Ivanov et al. [4], who restore cooperation through an incentive-compatible mediator. Under strictly rivalrous (all-or-nothing) contention, policy-level reciprocity has no efficient lever, and a centralized need-based allocator—which decides who receives each resource—is an upper bound on fairness that any decentralized method should aim to match [5]. We take that target seriously and pursue the decentralized side it leaves open.

Reciprocity and conditional cooperation. Inequity aversion [8] and learned reciprocity [9] make cooperation conditional on others' behaviour in social dilemmas. Our cooperators are likewise behaviour-conditioned, but the obstacle here is sharper: under all-or-nothing contention reciprocity has no efficient lever at all; we show graded contention restores one and that a learned attention policy can exploit it.

Attention in MARL. Attention over agents is well established for value/critic mixing and communication, e.g. actor-attention critics [10] and the centralized critics of MADDPG [11]. We use a single-head attention block [12] *in the decentralized policy*, each agent's query attending across all agents' observed-behaviour tokens (Sec. IV); its role is contention inference, and its permutation-equivariance gives a single N-agnostic policy that can be evaluated at unseen team sizes.

League / population training. Training against a growing population of past best responses—fictitious play [13] and PSRO [6]—yields policies robust to a history of adversaries rather than to a single co-evolving one. We use it to stabilize the cooperator against the worst free-rider it has seen, which we find is what makes robustness consistent across contention levels.

III. The Graded-Contention Game

Game. N agents act over T steps. Each step every agent chooses Claim (1, the self-interested act) or Yield (0). Let $\mathbf{u} \geq 0$ accumulate the resource each agent has collected and let m be the number of claimers this step. The unit resource is allocated by the *graded-contention* rule

$$g(m) = \begin{cases} 1 & m = 1, \\ (1-c)\,/\,m & m \geq 2, \\ \text{routed to the worst-off agent} & m = 0, \end{cases} \tag{1}$$

where each claimer receives $g(m)$ and $c \in [0, 1]$ is the contention *waste*. A sole claimer wins the whole unit; $m \geq 2$ claimers split a discounted $1-c$; if nobody claims, the resource flows (losslessly) to the worst-off agent. The parameter c interpolates from lossless sharing (c=0) to the all-or-nothing model (c=1, where contesting destroys the resource). One caveat is worth stating plainly: the m=0 branch is itself a need-based mechanism—when nobody claims, the *rule*, not an agent, routes the unit to the worst-off. Our no-central-allocator claim therefore concerns contested steps: whenever $m \geq 1$, who gets what is decided entirely by the agents' own Claim/Yield choices, and that is where the robustness problem lives.

Threat model. Each episode, each agent is independently a *defector* with some probability, so the number of free-riders D is variable and unknown (including D=0); defectors always claim. The remaining *cooperators* share one policy and seek to keep the allocation fair and efficient. We audit a trained cooperator policy by *freezing* it and training a fresh best-response defector to maximize its own utility—an exploitability test stronger than a fixed always-claim adversary.

Bounded free-ride metric. We measure exploitation by the free-ride factor of the defector *group*: its share of the delivered total, normalized by its fair share n_{def}/N,

$$\rho \;=\; \frac{N \sum_{i \in \mathcal{D}} u_i}{n_{\text{def}} \sum_j u_j} \;\in\; [0,\, N/n_{\text{def}}], \tag{2}$$

where $\mathcal{D}$ is the defector set. ρ=1 is exactly fair; ρ=N means a lone defector took everything. Eq. (2) is *bounded* by construction, unlike the naive ratio of defector utility to mean-cooperator utility, which diverges when cooperators are starved as $c \to 1$; the delivered total is always positive, so ρ stays in range and is comparable across c. We also report Jain's index [14] and efficiency (delivered/T).

Leverage. Under all-or-nothing contention (c=1) a cooperator cannot deny a free-rider except by wasteful levelling-down—contesting collides and destroys the unit—so a welfare-fair team is indifferent between yielding and contesting and has no efficient defense. Graded contention breaks that premise.

Proposition 1 (Decentralized leverage under graded contention): Fix $c < 1$ and a step at which a single free-rider claims. Let a worst-off cooperator choose Claim rather than Yield. Then the cooperator's received resource rises from 0 to $(1-c)/2 > 0$, while the free-rider's falls from 1 to $(1-c)/2$. Hence contesting *strictly dominates* yielding for the worst-off cooperator, and strictly reduces the free-rider's appropriated surplus.

Proof. If the worst-off cooperator yields, the free-rider is the sole claimer (m=1), receives $g(1)$=1, and the cooperator receives 0. If the cooperator contests, m=2 and each receives $g(2)$=$(1-c)/2$. Since $c < 1$, $(1-c)/2 > 0$, so the cooperator strictly gains and the free-rider strictly loses. □

The dominant action is also *objective-aligned*: lifting the worst-off cooperator from 0 to $(1-c)/2$ raises the cooperator mean and lowers their spread, so it strictly increases the training welfare W_{coop}=mean−std (Sec. IV). Contesting is thus rewarded both at the level of the individual worst-off agent and of the team objective.

Dominance, not equilibrium. Proposition 1 is a per-step dominance statement, not an equilibrium analysis of the repeated game: it fixes a step on which a single free-rider claims and gives the worst-off cooperator's myopic best reply. It does not assert that always-contesting is an equilibrium of the T-step game (it is not: when D=0, contesting burns c for nothing), nor that a defector's best response is to always claim—against a team that contests proportionally, each extra claim earns only the marginal share $(1-c)/m$. Empirically the two come apart: the *trained* best-response defector is deterrable—against a league-trained cooperator it converges to *end-game* claiming (claim rate ≈0 over the first three episode quarters, ≈0.65 in the last, at both c=0.5 and c=0.9), harvesting where the finite horizon leaves punishment no future to bite—whereas a *committed* always-claim script ignores punishment, and at high c it extracts more than the trained best response finds (Sec. V-E). Our audits therefore report the worse of the two. A full equilibrium characterization of the repeated graded-contention game is open; our claims are the existence of the per-step lever and the empirical exploitability audit built on it.

Proposition 1 establishes that the lever *exists* for every $c < 1$; it does not say how to pull it. The cooperator must contest *only* when a free-rider is present (else it pays the waste c for nothing) and only *enough* cooperators must contest (over-contesting wastes; under-contesting leaves surplus). Because D is unknown and variable, this is an inference-and-coordination problem, which motivates a behaviour-conditioned learned policy.

IV. Method: Cross-Attention Networks (CAN)

Observed-behaviour tokens. At step t each agent i is represented by a token of six features computable from public state—no access to others' intentions or rewards:

$$x_i = \left[\tfrac{u_i}{T},\ \tfrac{u_i-\bar{u}}{T},\ \tfrac{u_i-u_{\min}}{T},\ \mathbf{1}[u_i{=}u_{\min}],\ \tfrac{cc_i}{t},\ \tfrac{t}{T}\right], \quad (3)$$

where $\bar{u}, u_{\min}$ are the team mean/min utilities and cc_i/t is agent i's running claim-rate. The claim-rate channel is what lets a cooperator estimate *how many* others are grabbing.

Architecture. CAN's aggregator is a single-head attention block over the N tokens: with Q, K, V linear projections of $\{x_i\}$, $\text{ctx} = \text{softmax}(QK^\top/\sqrt{d})V$; each agent concatenates its token with its context, passes a tanh layer, and outputs CLAIM/YIELD logits (Fig. 1a). We use *cross-attention* in the inter-agent sense—agent i's query attends *across* all agents' behaviour tokens to build its context—computed for all agents at once as a single self-attention pass over the token matrix; it is not the two-sequence encoder–decoder block sometimes meant by the same term. The block is shared by all cooperators and *permutation-equivariant*, so (i) the policy is decentralized—agent i acts on the public tokens, and on every contested step ($m \geq 1$) no central allocator decides the allocation (the m=0 fallback is the mechanism's, Sec. III)—and (ii) it is N-agnostic, enabling zero-shot evaluation at unseen team sizes.

Cooperator objective. Cooperators maximize a per-step welfare reward-to-go with welfare

$$W_{\text{coop}} = \text{mean}_{i\in\mathcal{C}}(u_i) - \text{std}_{i\in\mathcal{C}}(u_i), \quad (4)$$

over cooperators $\mathcal{C}$: the mean term rewards reclaiming utility from a free-rider (contest when $D \geq 1$), the std term rewards turn-taking (share evenly), and doing nothing when D=0 avoids the waste c. Training is REINFORCE with an annealed entropy bonus; the defector(s) during training draw $D \sim \text{Unif}\{0, \ldots, d_{\max}\}$ per episode so the policy sees a variable, unknown adversary count.

Training schemes. We compare three ways to provide the adversary, holding the cooperator architecture fixed: *(i) single co-training*—cooperators and one co-evolving defector policy trained together; *(ii) population co-training*—a population of K=4 co-evolving defectors, one assigned per episode; *(iii) league (PSRO)* [6]—alternate training the cooperators against the frozen pool of all past best-response defectors and adding a fresh best response to the pool (Fig. 1b). League training targets the multi-equilibrium fragility of single co-training: the cooperator becomes robust to the whole adversary history rather than to one moving target.

Baselines and oracle. Two fixed scripts bracket the problem: *all-contest* (everyone always claims; ρ≈1 but efficiency $1-c$) and *yield* (cooperators always yield; efficiency 1 but ρ=N). The *centralized oracle* allocates by need (worst-off first), achieving ρ=1 at efficiency 1—the unbeatable upper bound a decentralized policy aims to approach.

V. Experiments

Setup. Unless noted, N=6, T=100, $d_{\max}$=2, $c \in \{0.3, 0.5, 0.7, 0.9\}$, 5 seeds; all policies are shared single-head cross-attention nets (hidden 64) trained with REINFORCE/Adam (lr 3×10^{-3}, B=512 parallel episodes). Every reported policy is audited by a freshly trained best-response defector (Eq. (2)). Implementation details are in the appendix.

A. Head-to-head with fair-MARL learners

We first place CAN against the welfare-fair learners that motivate the problem, *on the same graded-contention game and under the same bounded audit*: a shared-policy GGF maximizer, a faithful SOTO (Self-/Team-Oriented sub-nets, annealing β:1→0) [2], and a FEN-style learner whose agents maximize the fair-efficient reward $\bar{u}/(c_0+|u_i-\bar{u}|)$ [1]. All three see the same six behaviour features and are trained *cooperatively* (no defector at train time, as in their originals);

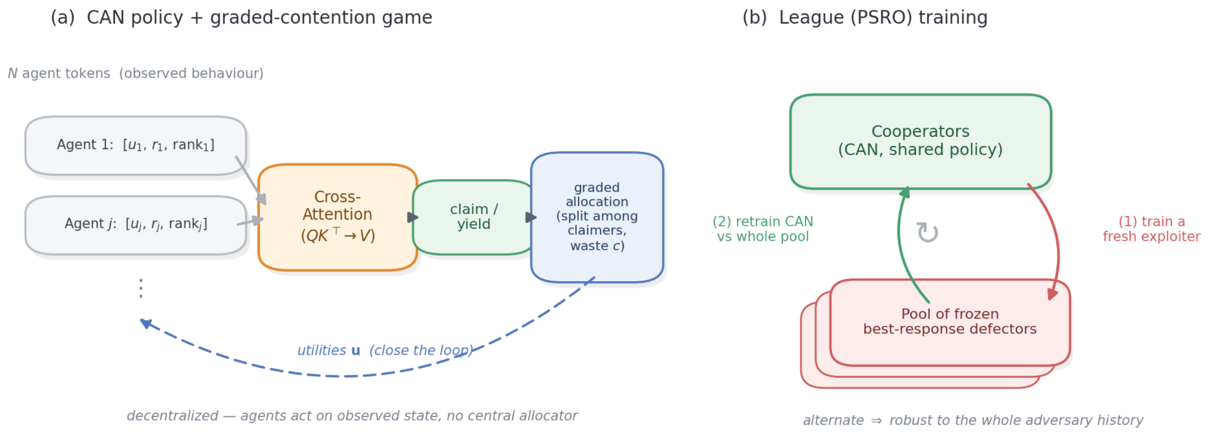

Fig. 1: Overview. (a) The CAN policy: each agent is a token of observed behaviour (utilities, claim-rate, who is worst-off); a shared cross-attention block infers contention and outputs claim/yield; the graded allocator returns utilities, closing a fully *decentralized* loop. (b) League (PSRO) training: alternate retraining the cooperators against the frozen pool of past best-response defectors and adding a fresh exploiter, yielding robustness to the whole adversary history.

TABLE I: Head-to-head on the graded-contention game (N=6, 5 seeds). Each fair learner fails on one axis; CAN attains both. Ranges span $c \in \{0.3, \ldots, 0.9\}$.

Method	D=0 eff.	best-resp. ρ	outcome
GGF (welfare)	1.00	6.0	efficient, exploitable
FEN	0.68–0.98	3.0–6.0	unstable, mostly exploitable
SOTO	$1-c$	1.0	robust, wasteful
CAN (league)	**0.98**	**1.2–1.5**	**both**
centralized oracle	1.00	1.0	(upper bound)

we then freeze each team and train a best-response defector. This comparison is therefore *fair-as-published vs. robust-by-design*: it isolates whether the fair *objective alone* confers robustness (it does not). The training axis is isolated separately: Sec. V-G varies the adversarial scheme on CAN's own architecture, and Sec. V-E league-trains the fair objectives themselves under CAN's identical budget—which does not rescue them.

The result is a clean two-axis separation (Fig. 2, Table I). Every fair learner is fair at D=0 (Jain ≈1.0), but each collapses onto one of the two scripted extremes and fails on one axis. **GGF** learns to *yield*: efficient (D=0 eff. 1.00) but maximally exploitable (ρ=6=N at every c—a lone defector takes everything). **SOTO** learns to *all-contest*: robust (ρ=1.0) but wasteful—its D=0 efficiency is exactly $1-c$ (down to 0.10 at c=0.9), because its "be selfish first, then anneal to fair" schedule settles in a fair-by-symmetry but resource-destroying equilibrium. **FEN** is unstable, landing in *either* bad corner depending on seed and c (at c=0.5 most seeds learn yield, $\rho \approx 5.9$, a minority all-contest, ρ=1.0, giving a high-variance mean); its fair-efficient reward does not reliably resolve the tradeoff. **CAN** is the only method in the good corner: efficient (0.98) *and* robust (ρ=1.2–1.5), clustered beside the centralized oracle. The remaining experiments isolate the two properties that put it there.

B. Efficiency when no free-rider is present

A robust cooperator must not pay for a defense it does not need. Fig. 3 shows the D=0 efficiency of CAN against

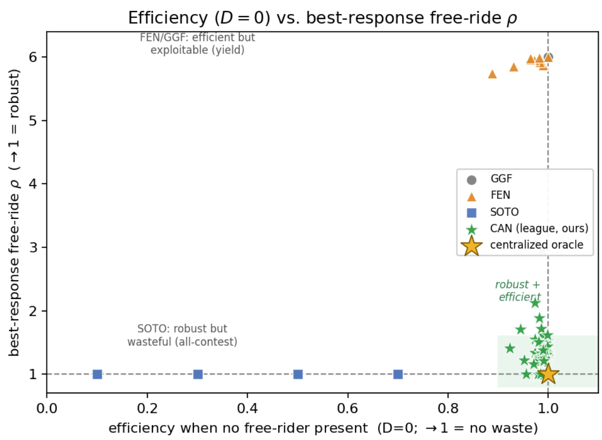

Fig. 2: Efficiency vs. exploitability (each marker is one seed×c). The welfare-fair learners collapse onto the scripted extremes—GGF/yield (efficient, ρ=N) and SOTO/all-contest (robust, eff. $1-c$)—and FEN scatters between them. Only CAN occupies the good corner (shaded), beside the centralized oracle.

the fixed all-contest script. CAN delivers 0.99–1.00 of the resource at *every* contention level, whereas all-contest delivers only $1-c$ (down to 0.10 at c=0.9). Because CAN turn-takes when it detects no free-rider, it incurs essentially no contention waste—matching the centralized oracle on efficiency. This is the half of the problem fixed rules cannot get right.

C. *Efficiency* under *a defector: is robustness free?*

D=0 is the easy case. The axis where a decentralized policy could lose to the centralized oracle—which delivers ρ=1 and efficiency 1 *simultaneously*—is efficiency *with* a free-rider present, since contesting forces $m \geq 2$ and delivers only $1-c$ on contested steps; in principle a policy could buy a low ρ at c=0.9 by contesting so hard that it wastes as much as the all-contest equilibrium it is meant to beat. It does not. Under the best-response defector, league-trained CAN keeps $D \geq 1$ efficiency at 0.88, 0.88, 0.83, 0.96 for c=0.3, 0.5, 0.7, 0.9 (pooled over 10 seed-runs)—at the hardest setting it delivers

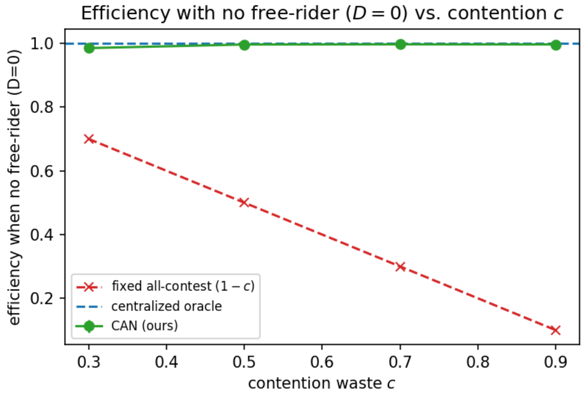

Fig. 3: Efficiency with no free-rider present (D=0). CAN turn-takes and wastes essentially nothing ($\approx$1.0) at all c, while a fixed all-contest policy pays the full waste $1-c$. Mean over 5 seeds.

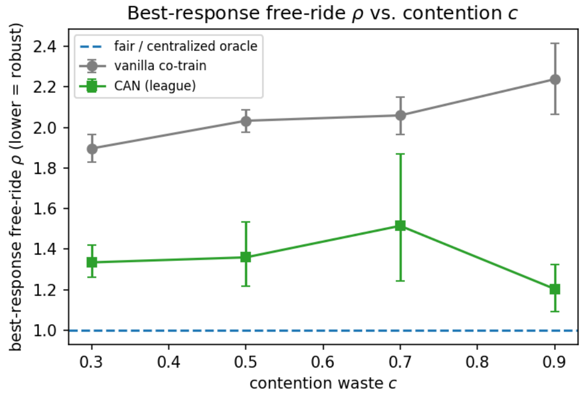

Fig. 4: Robustness vs. contention. League-trained CAN keeps ρ low (1.20–1.51) and is most robust at the highest contention, while the vanilla co-trained policy is roughly twice as exploitable and drifts upward. The fair / centralized-oracle line is ρ=1. Mean with 95% bootstrap CI, pooled over independent gen-6 runs (15 league / 10 vanilla seed-runs).

TABLE II: Best-response free-ride ρ vs. contention c (N=6, mean with 95% bootstrap CI; pooled over independent gen-6 runs—league 15, vanilla 10 seed-runs per c. Lower is better, 1 = fair, oracle = 1.0).

c	0.3	0.5	0.7	0.9
vanilla co-train	1.90	2.03	2.06	2.24
95% CI	[1.83,1.96]	[1.98,2.09]	[1.97,2.15]	[2.07,2.42]
CAN (league)	1.33	1.36	1.51	**1.20**
95% CI	[1.26,1.42]	[1.22,1.53]	[1.25,1.85]	[1.10,1.32]

0.96 versus the all-contest equilibrium's $1-c$=0.10. CAN contests *selectively* (just enough, and mostly via turn-taking rather than collisions), so its robustness is essentially free: it sits near the oracle on *both* axes even with a defector present, where SOTO reaches ρ=1 only by destroying 90% of the resource. A residual gap to the oracle remains (efficiency < 1), as expected for a decentralized policy, but it is small, not the structural $1-c$ collapse.

D. Robustness across the contention range

Fig. 4 and Table II report best-response exploitability ρ vs. c for CAN trained with vanilla co-training (against a fixed always-claim adversary) versus league (PSRO) training (against a growing pool of learned exploiters; to average over run-to-run variance we pool three independent gen-6 league runs, 15 seed-runs per c). League training keeps ρ low across the entire range (1.20–1.51), far below the vanilla co-trained policy (1.90–2.24) and well below the unprotected ρ=N=6. It is in fact *most* robust at the highest contention (ρ=1.20 at c=0.9, the lowest point)—consistent with leverage: at higher waste a single contest caps the free-rider's per-step take at $(1-c)/2$, so targeted contention denies it more effectively (Prop. 1). Vanilla co-training shows the opposite drift, rising to 2.24: training against a static adversary does not teach *when* to contest. Training against a *learned* exploiter is what converts the existence of leverage (Prop. 1) into robustness. One qualification is established in Sec. V-E: these ρ values audit with a *trained* best-response defector, which turns out to be a deterrable adversary; a *committed* always-claim defector extracts more at high c, where the trained-defector audit is most flattering.

E. Hardening the fair objectives, and a committed-defector audit

The head-to-head of Sec. V-A compared fair-as-published against robust-by-design, leaving open whether the fair objectives could be rescued by adversarial training alone. We close that cell: GGF, FEN, and SOTO are wrapped in the *identical* league loop as CAN—same tokens, same $6 \times (1200{+}1000)$ update budget, same per-step reward-to-go estimator with entropy annealing, same $D \sim \text{Unif}\{0..2\}$ training distribution—with only the welfare function swapped (each reduces to its published objective at D=0; SOTO's β anneals globally over the league's cooperator updates) and the per-agent MLP architecture of the originals. Every audit reports the *worse* of two adversaries: the standard trained best response and a scripted always-claim defector (5 seeds per cell; Table III, Fig. 5).

Hardening does not rescue the fair objectives. FEN+PSRO has exactly two attractors: all-contest at $c \leq 0.7$ (D=0 efficiency exactly $1-c$, ρ≈1) and, at c=0.9, a per-seed bifurcation between yield (efficient but ρ≈6=N *after six league generations*) and all-contest (efficiency 0.10). GGF+PSRO and SOTO+PSRO stay efficient (0.90–0.99) and partially deter the *trained* exploiter (ρ≈1.4–2.3), but the committed script collects ρ≈2–4.8 as c rises. No method reaches the good corner (D=0 eff. $\geq$ 0.95 and $\rho \leq 1.6$) beyond c=0.3. Adversarial training alone is therefore *not* the mechanism behind CAN's position in Fig. 2: the welfare-fair objectives are structurally hard to harden.

The dual audit also sharpens CAN's own limit. Retraining CAN's league under the dual audit reproduces the trained-best-response ρ of Table II within CIs, but the committed always-claim defector extracts 1.28 / 1.72 / 2.29 / 3.93 at c=0.3–0.9. The interpretation of Table II's flat ρ therefore changes: league training produces *deterrence*—the trained exploiter is punished into hiding, converging to end-game claiming (claim rate ≈0/0/0/0.65 by episode quarter), the finite-horizon shape of a deterred adversary—rather than *immunity*. A committed defector ignores punishment, and sustaining the threat costs the contesters $(1-c)/2 \to 0$,

TABLE III: Hardened baselines vs. CAN under the dual audit: $D{=}0$ efficiency / committed-defector exploitability $\max(\rho_{\mathrm{BR}}, \rho_{\mathrm{always}})$, mean over 5 seeds. No hardened fair objective reaches the good corner; CAN is best everywhere but decays with c.

c	0.3	0.5	0.7	0.9
GGF+PSRO	0.94 / 1.60	0.98 / 2.04	0.99 / 2.88	0.99 / 4.76
FEN+PSRO	0.70 / 1.04	0.50 / 1.02	0.37 / 1.78	0.63 / 4.07
SOTO+PSRO	0.90 / 1.55	0.98 / 2.02	0.97 / 2.50	0.99 / 4.26
CAN (league)	**0.97 / 1.28**	**0.97 / 1.72**	**0.99 / 2.29**	**1.00 / 3.93**

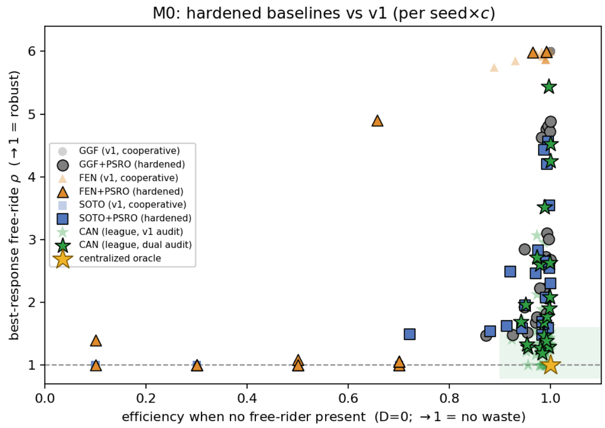

Fig. 5: Hardened baselines vs. v1 on the efficiency–exploitability plane (per seed$\times c$; hardened points outlined, v1 cooperative training faded). League-training the fair objectives moves FEN/SOTO toward the wasteful all-contest corner and leaves GGF efficient but exploitable; no hardened baseline reaches the good corner. CAN under the dual audit (committed always-claim included) remains the least exploitable efficient policy but decays toward high c—deterrence, not immunity.

so robustness to commitment decays exactly as Prop. 1's leverage vanishes. CAN remains the least exploitable efficient policy at every c under the stricter score, but the deterrence-vs-commitment gap grows with c for every method, CAN included. This is the honest empirical match to the theory: decentralized defense works *as far as leverage reaches*, and no farther.

F. Larger $d_{\max}$ and defector coalitions

The main results cap the per-episode defector count at $d_{\max}{=}2$, an easy "0/1/2 grabbers" inference. We stress this by training CAN with $d_{\max}{=}4$ (up to four of six defecting) and auditing against a best-response *coalition* of D defectors sharing one policy. CAN remains robust at all c and D (Table IV): the worst case is a *lone* free-rider ($\rho{=}1.13$–1.60, comparable to the $d_{\max}{=}2$ results), and larger coalitions are *easier*—$\rho{\to}1.0$ for $D{\ge}2$—because with more agents defecting there is less cooperative surplus to appropriate. The harder inference and coordinated free-riders thus do not break the policy.

G. Choice of adversarial training scheme

Does the result depend on the *league* specifically? Fig. 6 compares three adversarial schemes—single co-training (one co-evolving defector), population co-training ($K{=}4$ defectors), and the league—at the two lower contention levels, per seed. The schemes are statistically indistinguishable: at $c{=}0.3$ all three average $\rho{\approx}1.48$, and at $c{=}0.5$ they span 1.18–1.46 with overlapping CIs. What carries the result is *that* the

TABLE IV: Larger-$d_{\max}$ stress: best-response ρ for CAN trained at $d_{\max}{=}4$, audited against a D-defector coalition ($N{=}6$, 5 seeds).

c	$D{=}1$	$D{=}2$	$D{=}3$
0.3	1.22	1.11	1.07
0.5	1.13	1.00	1.00
0.7	1.51	1.00	1.00
0.9	1.60	1.05	1.00

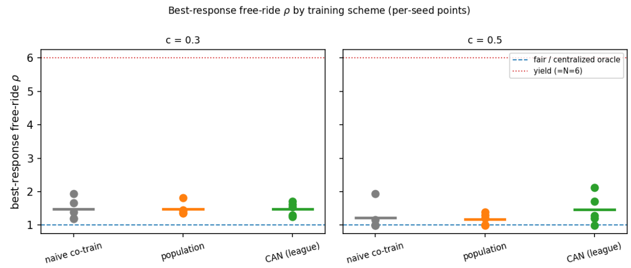

Fig. 6: Per-seed best-response ρ for three adversarial training schemes (5 seeds). The schemes are indistinguishable (all $\rho{\approx}1.2$–1.5); what matters is training against a *learned* exploiter rather than a static one. The dotted line marks the all-yield ceiling ($\rho{=}N$).

cooperator is trained against a *learned*, adapting exploiter at all: every adversarial scheme lands near $\rho{=}1.2$–1.5, versus 1.8–2.2 for the static-adversary vanilla baseline (Table II). We adopt the league for its robustness to the whole adversary history, but the headline robustness is not an artifact of that choice.

H. What is load-bearing? Architecture ablation

Is the win the attention, or just behaviour-conditioning with *some* adaptive aggregation? We test this by replacing the cross-attention block with three alternative aggregators of the *same* behaviour tokens, under *identical* league training and the same best-response audit: a *mean-pool* policy (each agent conditions on its token and the mean of all tokens), a *deep-sets* policy (mean-pool of a learned per-agent embedding), and a bidirectional *GRU* over the agent tokens. Table V reports best-response ρ (mean over c, 5 seeds).

The clear conclusion is that *aggregation matters but pooling is not enough*: the mean-pool and deep-sets policies are markedly more exploitable than CAN ($\rho{=}1.78, 1.75$ vs. 1.37), so the natural objection "a permutation-equivariant pool should already match attention" does not hold. The gap to the strongest alternative is narrower and we do not overstate it: the bidirectional GRU is competitive on the mean (1.56 vs. 1.37) and even better at low c, so we credit *behaviour-conditioning plus adaptive aggregation*, not attention per se. Where cross-attention earns its place is *stability*: at the hardest contention ($c{=}0.9$) CAN holds $\rho{=}1.16{\pm}0.20$ while the GRU is erratic ($1.99{\pm}1.50$), so attention is the most stable instance of the family rather than categorically superior. These comparisons are at the deployed training budget; we found that under a *lighter* budget the CAN–GRU ranking reverses, underscoring that architecture ablations must use the budget at which the method is actually trained.

TABLE V: Architecture ablation under identical league training: best-response ρ (mean over c, 5 seeds; lower is better). Pooling policies (†, run at a lighter budget) are more exploitable than CAN *even when CAN is held to that same lighter budget* (ρ=1.57); at full budget CAN improves to 1.37 and also beats the bi-GRU, chiefly by staying stable at the hardest contention. CAN's ρ in this ablation (1.37) is one of the three independent gen-6 league runs pooled in Table II (pooled mean 1.35); all audit with a fresh best-response defector, and the spread reflects run-to-run (GPU) variance.

Policy (aggregator)	best-resp. ρ	ρ at c=0.9
mean-pool†	1.78	2.05
deep-sets†	1.75	2.06
bidirectional GRU	1.56	1.99±1.50
cross-attention (CAN)	**1.37**	**1.16±0.20**

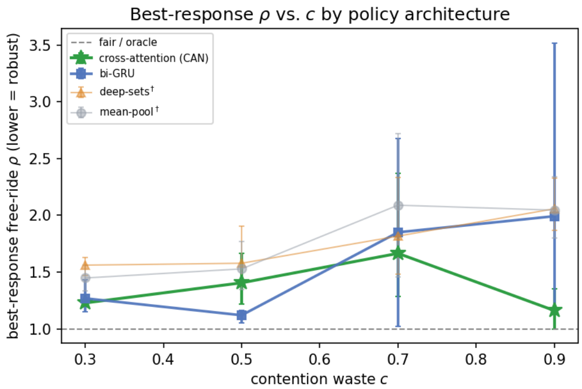

Fig. 7: Architecture ablation: best-response ρ vs. c by aggregator (5 seeds, 95% bootstrap CI). Cross-attention is the only policy that stays low *and* becomes most robust at the hardest contention (c=0.9); the bi-GRU is competitive at low c but erratic at high c, and the pooling baselines (†, lighter budget) are uniformly worse.

I. Zero-shot transfer across team size (a limitation)

Because CAN is permutation-equivariant it can be evaluated at team sizes it never trained on. Fig. 10 applies the N=6 policy to teams of 12 and 24. Transfer holds well at low contention—at c=0.3 a 4× larger team is still close to fair (ρ=1.85 at N=24)—but degrades at high contention, reaching ρ=8.8 at N=24, c=0.9. The bounded metric keeps these readable (all curves stay below the all-yield ceiling ρ=N), and the ordering is clean: difficulty grows with both team size and waste. We report this as a genuine fragility—the contention-inference learned at N=6 does not fully generalize to crowds at high waste—rather than smoothing it over. The obvious remedy does not work: training on a size curriculum ($N \in \{4, 6, 8, 12\}$, evaluating at the held-out N=24) helps only at c=0.3 and *worsens* mid-range transfer (it dilutes the per-size budget), leaving the high-contention gap unchanged (ρ=4.5 at c=0.9, vs. 4.5 for N=6-only).

Two follow-up experiments locate the failure precisely, and it is *not* team size per se (Fig. 8). First, re-parameterizing the tokens to be N-invariant in scale (utility *shares* and z-scored deviations in place of T-normalized levels, plus explicit claimed-fraction channels) leaves the high-c gap unchanged (ρ=9.2 at N=24, c=0.9), and the trained policy's attention stays proportionally sharp at every team size—so neither input scale drift nor attention dilution is the cause. Second, evaluating at a constant defector *fraction* (D=$N/6$ rather than a lone defector), the unmodified N=6 policy transfers essentially perfectly: ρ=2.5 → 1.5 from N=6 to N=48 at c=0.9, flat or improving at every c (dashed curves in Fig. 8). What degrades with N is the *detection of a single defector*: a lone free-rider is a vanishing fraction of every aggregate the policy can form (its weight under any convex pooling scales $\sim 1/N$), so the onset of contestation arrives later as N grows—at N=48, c=0.9 cooperators are nearly silent for the first $\sim$30 steps, during which the sole-claiming defector banks full, waste-free units. The free-ride against a lone defector is a *count* quantity, not a fraction quantity, and no mean-field-style parameterization preserves it. Breaking the $1/N$ scaling is partially possible (Fig. 9): adding a count-preserving max-pooled suspicion branch to the aggregation, fed by an instantaneous claimed-last-step channel, restores near-immediate detection at N=48 and roughly halves the lone-defector ρ twice over (N=24, c=0.9: 8.9 → 4.6; N=48: 20.2 → 6.9; the channel is useless without the max statistic and vice versa at high c). What remains is a *structural floor*: with detection latency L steps, the defector banks L full units while every later contested step delivers only $1-c$, so $\rho \approx NL/(L+(T-L)(1-c))$ stays high as $c \to 1$ even under near-perfect detection—the price of the detection window, not an inference failure.

J. Generality across environments

The experiments so far use the single-resource graded game. To test whether CAN's behaviour-conditioned contention inference is specific to that game, we replicate the head-to-head on two further leverage-preserving environments that pose *different* inference problems (Fig. 11, Table VI). *(i) Congestion*: M=3 parallel servers; each agent claims one or yields; equal-split per server; empty servers route to the neediest. The cooperators must infer *which* server each free-rider targets and contest just that one. *(ii) Stakes*: a single resource whose value v_t is random (lumpy: usually small, occasionally a jackpot, $\mathbb{E}[v]$=1). The cooperators must contest *selectively*, on the high-stakes steps where a free-rider's surplus concentrates. Both keep equal-split contention, so Prop. 1 leverage holds; CAN and the three welfare-fair learners are trained and audited exactly as before.

The efficiency axis is what carries over. On exploitability alone the games differ in difficulty (the congestion exploit is structurally capped at ρ=N/M=2; stakes restores the ρ=N ceiling). The one property that is consistent is efficiency: **CAN keeps** D=0 **efficiency** 0.94–1.00 **in all three games**, so it never buys robustness by wasting the resource—unlike the only baseline that attains low ρ, SOTO, whose all-contesting collapses efficiency to 0.40–0.47 (and to 0.10 at stakes c=0.9). GGF and FEN are Pareto-dominated by CAN in every game. Robustness, however, does *not* carry over uniformly: CAN is both efficient and robust on base and congestion, but on stakes it is only *partially* robust—robust up to moderate contention and exploited at c=0.9—so we claim a clear scope, not blanket generality.

An honest gradient: leverage strength governs success. CAN is not uniformly robust; its exploitability tracks how much leverage Prop. 1 actually delivers. With *strong* leverage (base: a contest caps the free-rider at $(1-c)/2$ for any c) CAN is robust at every c and *most* robust at the highest contention (ρ=1.20 at c=0.9). With *structurally capped*

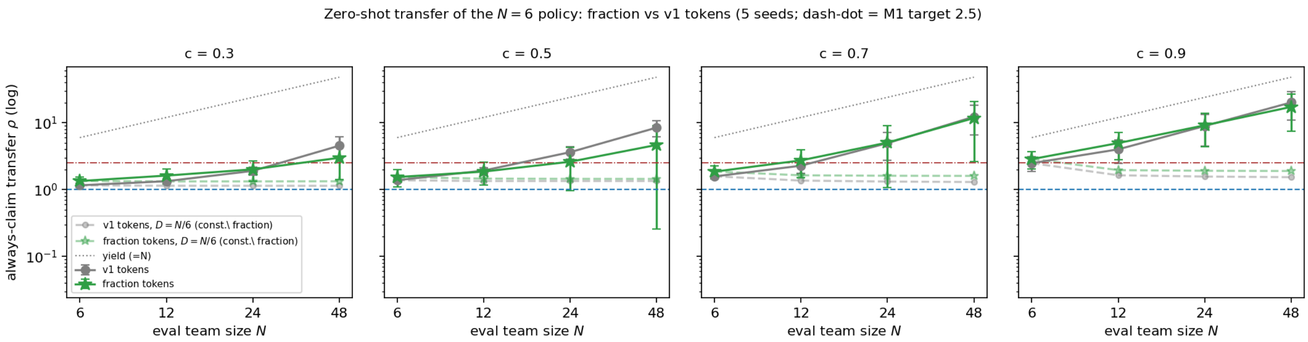

Fig. 8: The transfer fragility is the *lone* defector, not team size. Solid: lone-defector (D=1) zero-shot transfer of the N=6 policy, v1 vs N-invariant fraction tokens—both degrade toward the all-yield line as N grows at high c. Dashed: the same policies at constant defector *fraction* (D=$N/6$) transfer essentially perfectly at every c. Log–log; dash-dot line marks ρ=2.5.

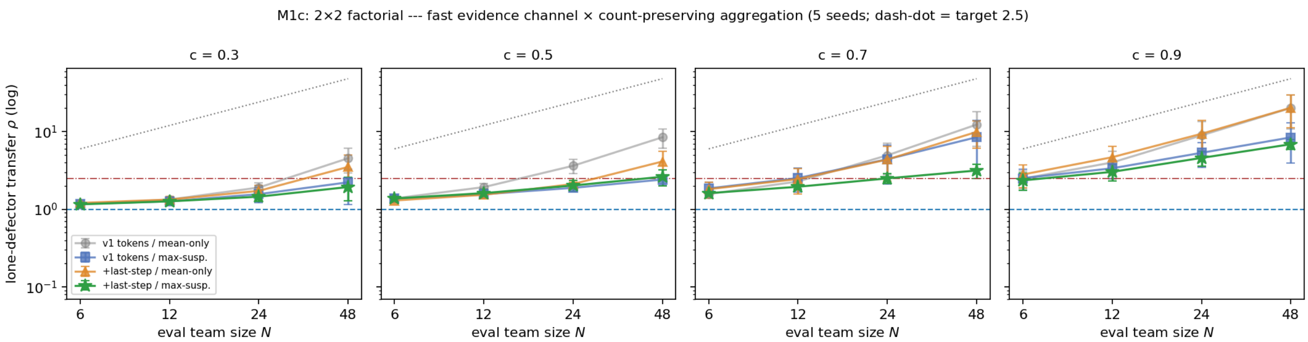

Fig. 9: Breaking the $1/N$ evidence scaling (2×2 factorial). The instantaneous claimed-last-step channel alone (△) matches the v1 control—fast evidence is useless under convex (mean-style) pooling; the count-preserving max-suspicion branch alone (□) halves the lone-defector gap; their combination (⋆) halves it again and meets the ρ<2.5 target through c=0.7. The c=0.9 residual is the structural floor $\rho \approx NL/(L+(T-L)(1-c))$.

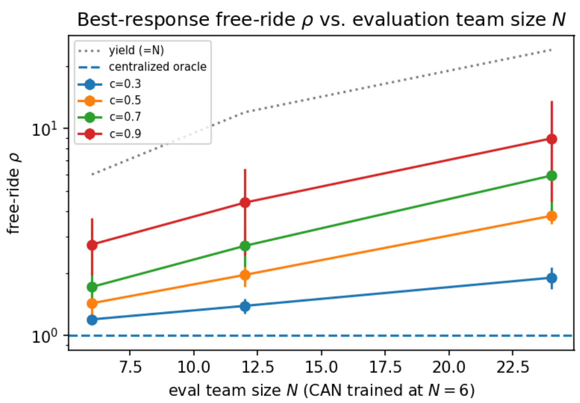

Fig. 10: Zero-shot transfer of the N=6 policy to larger teams (log scale). Robust at low contention; degrades with team size at high contention. All curves remain below the all-yield ceiling ρ=N; the oracle line is ρ=1.

leverage (congestion) it is robust but cannot beat the ρ=2 ceiling by much at high c. With *weakening* leverage (stakes: contesting a jackpot at c=0.9 returns only $0.05\,v$) it degrades to ρ=4.05 at c=0.9—while still staying efficient (0.97) where the robust alternative (SOTO) delivers only 0.10. And with *absent* leverage—a rich-get-richer (Matthew) rule where the *richest* claimer wins, so a worst-off contester captures nothing and Prop. 1 fails by construction—CAN is exploited (ρ≈5.5), marking the boundary of the approach. The method works as far as the leverage it is built on, and no further.

TABLE VI: Generality: (D=0 efficiency, best-response ρ) per method, mean over c (5 seeds). CAN is efficient (≥ 0.94) in every environment and the least exploitable efficient policy; SOTO's robustness costs catastrophic efficiency. $\rho_{\max}$ is the yield ceiling.

	base ($\rho_{\max}$=6)		congestion (2)		stakes (6)	
Method	eff	ρ	eff	ρ	eff	ρ
GGF	1.00	6.00	1.00	2.01	0.98	5.75
FEN	0.81	4.20	0.96	2.05	0.70	3.27
SOTO	0.40	1.00	1.00	2.01	0.47	1.64
CAN	**0.98**	**1.35**	**1.00**	**1.60**	**0.94**	**2.32**

VI. Discussion and Limitations

What this shows. The decentralization gap left open by prior work [5] is not fundamental: on the graded game a decentralized cross-attention policy trained against an adversarial league recovers most of the centralized oracle's incentive-compatibility against adaptive exploiters (ρ≈1.2–1.5; committed defectors are deterred only as far as leverage reaches, Sec. V-E) *and* its efficiency (≈1.0 when no free-rider is present), with no central allocator. The all-or-nothing futility result is the boundary case c=1. The effect is not automatic for every graded rule, though: it requires that contesting a free-rider actually return a positive share to the contester (Prop. 1), which the stakes and Matthew games show can be weak or absent.

Where it is fragile. (i) Zero-shot transfer against a *lone* defector degrades with team size at high contention (Fig. 10); neither a size curriculum nor N-invariant tokens fix it,

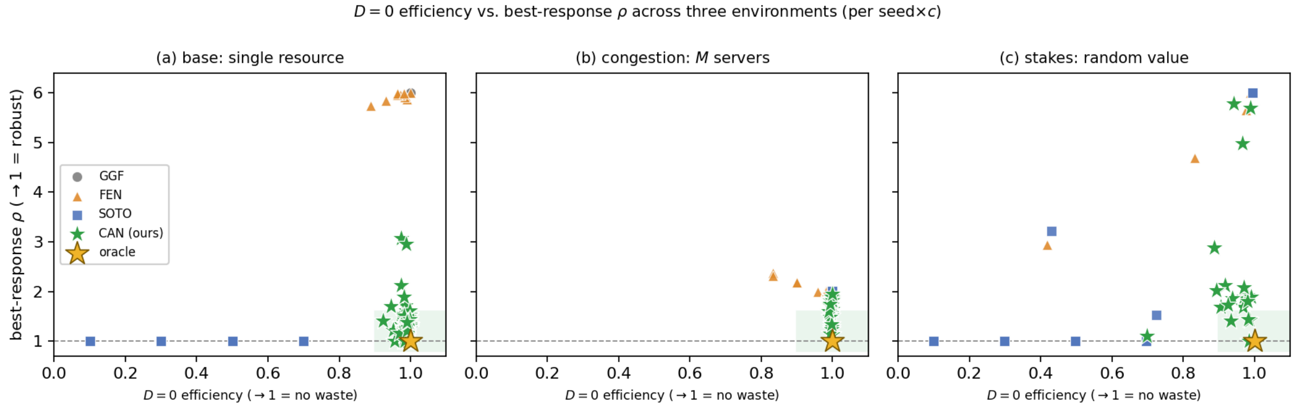

Fig. 11: D=0 efficiency vs. best-response ρ on three structurally distinct leverage-preserving games (each marker is one seed×c). In every game CAN (green) stays on the efficient (right) side while remaining the least exploitable efficient policy; the only robust baseline, SOTO (blue), reaches low ρ only by moving left into the wasteful region. GGF/FEN are efficient but exploitable (top-right). Shaded = the efficient-and-robust corner; gold star = centralized oracle.

because the cause is detection latency—a single defector is a vanishing fraction of every aggregate, so evidence scales $\sim 1/N$. Transfer at constant defector fraction already works; crowd-scale detection of a lone free-rider at high waste is the open problem. (ii) The single-adversary scheme is seed-sensitive at low contention; league training is needed for consistency, at extra training cost. (iii) Robustness is bounded by leverage: on the stakes game it degrades at high contention (where contesting a jackpot returns almost nothing), and under a winner-take-all (Matthew) rule it disappears entirely—CAN is robust only as far as Prop. 1's leverage extends. (iv) What league training buys is *deterrence* of adaptive exploiters, not immunity: a *committed* always-claim defector, which no amount of punishment dissuades, extracts more as $c \to 1$ (Sec. V-E)—for every method, ours included. In boundary experiments at c=1 the obstacle is *credibility*, not detection: scripted trigger policies over the same public signals (grim / tit-for-tat on observed claim rates) hold best-response and patient learning defectors at exactly ρ=1.0 at zero efficiency cost, while league-learned policies, whose punishment is probabilistic rather than absorbing, leave $\rho \approx 1.8$. This does not mean one should simply deploy the trigger: the scripts are hand-designed around a known threat model (a claim-rate flag with a tuned threshold) and are correct only at the boundary—where punishment is free because the contested unit is destroyed anyway—whereas at $c < 1$ permanent punishment burns surplus that proportional contesting would recover, which is exactly the regime the learned policy handles. The open problem is *learned, transferable* commitment: a single policy that acquires the trigger's credibility where leverage is absent without inheriting its rigidity where leverage exists. That is the subject of companion work; code for those experiments is included in the repository. (v) Our games are controlled abstractions with a single shared cooperator policy; spatial/continuous environments are still open (coordinated defector coalitions, by contrast, we do test, and they are no harder than a lone free-rider). (vi) The oracle remains an upper bound CAN approaches but does not reach.

Takeaway. Fair-MARL methods are usually reported on the fairness they achieve under full cooperation. We argue they should also be reported on the fairness they *retain* under self-interest—and that, contrary to the all-or-nothing intuition, a decentralized policy can retain much of it whenever the contention rule leaves a contester a real share to capture.

VII. Conclusion

We gave a decentralized account of robust fairness in cooperative MARL. Graded contention makes contesting a free-rider strictly worthwhile for the worst-off (Prop. 1), but exploiting this requires inferring an unknown, variable number of free-riders and responding proportionally. CAN—a permutation-equivariant cross-attention policy over observed behaviour, trained against an adversarial league—does so, holding trained-best-response exploitability low across the contention range while wasting essentially nothing when no free-rider is present, thereby approaching a centralized need-based oracle without a central allocator; against a committed defector this protection is a deterrence effect that holds in proportion to the remaining leverage. We are explicit about scope rather than claiming blanket generality: across additional games CAN stays efficient and dominates the welfare-fair learners, but its robustness holds in proportion to the contest leverage—strong on a multi-server game, only partial where the leverage weakens (high-value steps at high contention), and absent under a winner-take-all rule. We hope this reframes decentralized robust fairness as a tractable, measurable target—bounded by a clear condition—rather than an impossibility.

Code Availability

All code (the graded-contention game, CAN, the architecture and training-scheme ablations, and all experiments) and the LaTeX source are available at https://github.com/highcansavci/can-fair-marl.

Use of AI-Assisted Technologies

The author used Claude (Anthropic), including Claude Code, to assist with the experimental code, manuscript review and revision, and formatting. All AI-assisted content was verified by the author, who is solely responsible for the methodology, results, and conclusions of this paper.

Appendix A
Environment Details

All environments share a skeleton: N=6 agents act for T=100 steps; at each step divisible resource is available; each agent chooses a self-interested act (CLAIM) or to YIELD; accumulated utilities $\mathbf{u}$ determine fairness. They differ only in the *allocation rule* mapping the joint action to utility increments, and (for stakes/congestion) in the action space and one extra observation. Let m be the number of claimers of a resource and $c \in [0, 1]$ the contention waste.

Base (single-resource graded contention). One resource per step, binary action. Each claimer receives

$$g(m) = \begin{cases} 1 & m{=}1, \\ (1{-}c)/m & m \geq 2, \\ \text{(unit to worst-off } \arg\min_i u_i) & m{=}0, \end{cases}$$

so a sole claimer wins the unit, $m \geq 2$ claimers split a discounted $1{-}c$, and an unclaimed unit flows losslessly to the neediest agent. c=1 recovers the all-or-nothing model.

Congestion (multi-server). M=3 parallel unit resources ("servers") per step; action space $\{\text{YIELD}, \text{claim } 1, \ldots, \text{claim } M\}$ (M+1 actions). Each server is allocated independently by the base rule (k claimers $\rightarrow$ 1 if k=1, $(1{-}c)/k$ if $k \geq 2$); the E empty servers route to the E lowest-utility agents (one unit each, no waste). A lone free-rider can monopolise only one of M servers, so the exploit ceiling is $\rho{=}N/M{=}2$. Each token is augmented with the agent's per-server running claim-rates (so the policy can infer *which* server is being grabbed): 5+M features, (M+1)-way head.

Stakes (stochastic value). One resource per step with random value v_t drawn i.i.d. as v_t=3.25 w.p. 0.25 and v_t=0.25 otherwise (lumpy jackpots, $\mathbb{E}[v_t]$=1, so efficiency normalises as in base). The base rule scales by v_t: a sole claimer gets v_t, $m \geq 2$ claimers get $(1{-}c)v_t/m$ each, an unclaimed unit gives v_t to the worst-off. The current v_t is appended to each token (7 features), so the policy can contest selectively on high-value steps.

Matthew (rich-get-richer; boundary). One resource, binary action, but among claimers the *richest* wins: m=1 gives the sole claimer 1; $m \geq 2$ gives the richest claimer $(1{-}c)$ and the rest 0 (ties broken uniformly at random); m=0 routes to the worst-off. A worst-off cooperator that contests a richer free-rider receives 0, so Prop. 1 leverage vanishes; we use this environment only to mark the boundary of the method.

Threat model and metric (all environments). Each episode, D agents are *defectors* with $D \sim \text{Unif}\{0, \ldots, d_{\max}\}$ ($d_{\max}$=2 unless noted; the stress study uses $d_{\max}$=4); defectors always take the self-interested act and the $N{-}D$ cooperators share one policy. To audit a frozen cooperator policy we train a best-response defector (or a D-agent coalition sharing one policy) to maximise the defector group's utility, and report the bounded free-ride factor

$$\rho = \frac{N \sum_{i \in \mathcal{D}} u_i}{n_{\text{def}} \sum_j u_j} \in [0,\ N/n_{\text{def}}],$$

ρ=1 exactly fair, ρ=N a lone defector taking everything. Cooperators maximise the per-step increment of $W_{\text{coop}}{=}\text{mean}_{i \in \mathcal{C}}(u_i){-}\text{std}_{i \in \mathcal{C}}(u_i)$.

Appendix B
Implementation Details

CAN and all adversaries are shared single-head attention blocks (the inter-agent cross-attention of Sec. IV, computed as one self-attention pass over the $N{\times}6$ token matrix; hidden width 64): Q, K, V linear maps, scaled-dot-product attention, a $\tanh$ layer on $[\,x_i \,\|\, \text{ctx}_i\,]$, and a 2-logit head. All policies train with REINFORCE and Adam (lr 3×10^{-3}), B=512 parallel episodes, episodes of T=100 steps, with the entropy coefficient annealed from 0.05 to 0.003. Per-episode defector count is $D \sim \text{Unif}\{0, 1, 2\}$. Vanilla/population co-training run 3000 updates; league training runs 6 generations of 1200 cooperator updates against the pool and 1000 updates per fresh best response. Every audit trains a best-response defector for 1500 updates against the frozen cooperator and reports ρ from Eq. (2); the dual audit of Sec. V-E additionally rolls out a scripted always-claim defector at a random index and reports the worse of the two ρ values per seed. The hardened baselines of Sec. V-E reuse this league setup verbatim with the cooperator welfare replaced by the respective published objective masked to cooperators (GGF over cooperator utilities; FEN's fair-efficient reward with the mean over cooperators; SOTO's self/team sub-nets with β:1$\rightarrow$0 annealed globally over the 6×1200 cooperator updates) on the per-agent MLP (hidden 64) of the cooperative baselines. The cooperator welfare is the masked $\text{mean}{-}\text{std}$ over cooperators; advantages are discounted (γ=0.99) per-step welfare increments with a batch-mean (or learned value) baseline. The architecture ablation reuses this league setup with the cross-attention block replaced by a mean-pool, deep-sets, or bi-GRU aggregator; the mixed-N curriculum cycles the team size over $\{4, 6, 8, 12\}$ per update with the same total step budget. Unless noted, all numbers average 5 seeds and error bars are 95% bootstrap confidence intervals. League training is mildly sensitive to GPU non-determinism, which compounds over generations; the headline league/vanilla robustness (Table II, Fig. 4) therefore pools several independent gen-6 runs (15 and 10 seed-runs) so the CIs absorb run-to-run as well as seed variance.

References

[1] J. Jiang and Z. Lu, "Learning fairness in multi-agent systems," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2019.

[2] M. Zimmer, C. Glanois, U. Siddique, and P. Weng, "Learning fair policies in decentralized cooperative multi-agent reinforcement learning," in *International Conference on Machine Learning (ICML)*, 2021.

[3] U. Siddique, P. Weng, and M. Zimmer, "Learning fair policies in multi-objective (deep) reinforcement learning with average and discounted rewards," *International Conference on Machine Learning (ICML)*, 2020.

[4] D. Ivanov, I. Zisman, and K. Chernyshev, "Mediated multi-agent reinforcement learning," in *International Conference on Autonomous Agents and Multiagent Systems (AAMAS)*, 2023, pp. 49–57.

[5] C. Savcı, "On the exploitability of fair multi-agent reinforcement learning," *arXiv preprint*, 2026.

[6] M. Lanctot, V. Zambaldi, A. Gruslys, A. Lazaridou, K. Tuyls, J. Pérolat, D. Silver, and T. Graepel, "A unified game-theoretic approach to multiagent reinforcement learning," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2017.

[7] J. Z. Leibo, V. Zambaldi, M. Lanctot, J. Marecki, and T. Graepel, "Multi-agent reinforcement learning in sequential social dilemmas," in *International Conference on Autonomous Agents and Multiagent Systems (AAMAS)*, 2017.

[8] E. Hughes, J. Z. Leibo, M. Phillips, K. Tuyls, E. Duéñez-Guzmán, A. G. Castañeda, I. Dunning, T. Zhu, K. McKee, R. Koster *et al.*, "Inequity aversion improves cooperation in intertemporal social dilemmas," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2018.

[9] T. Eccles, E. Hughes, J. Kramár, S. Wheelwright, and J. Z. Leibo, "Learning reciprocity in complex sequential social dilemmas," *arXiv preprint arXiv:1903.08082*, 2019.

[10] S. Iqbal and F. Sha, "Actor-attention-critic for multi-agent reinforcement learning," in *International Conference on Machine Learning (ICML)*, 2019.

[11] R. Lowe, Y. Wu, A. Tamar, J. Harb, P. Abbeel, and I. Mordatch, "Multi-agent actor-critic for mixed cooperative-competitive environments," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2017.

[12] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, Ł. Kaiser, and I. Polosukhin, "Attention is all you need," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2017.

[13] J. Heinrich, M. Lanctot, and D. Silver, "Fictitious self-play in extensive-form games," in *International Conference on Machine Learning (ICML)*, 2015.

[14] R. K. Jain, D.-M. W. Chiu, and W. R. Hawe, "A quantitative measure of fairness and discrimination for resource allocation in shared computer systems," *DEC Research Report TR-301*, 1984.